FULL NAME: ______________________ **BLOCK:** __________

Real + Complex Zeros, and Modeling

Lesson 3-5 · Period 3 · Teacher Edition

OBJECTIVES

Math Objective	Find all zeros (real and complex) of a polynomial function, and use the factored form to model a volume-constrained application.
Language Objective	Students will use the frame "The zeros are ___; because ___, the complex conjugate ___ must also be a zero." to justify complex-zero reasoning.
Essential Understanding	Complex zeros of a polynomial with real coefficients come in conjugate pairs. A cubic model of a physical quantity (volume, profit) can be solved for specific values by factoring and solving.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students find all zeros (real + complex) of two cubics, then model a Storage Box volume problem end-to-end. Every item is Savvas-sourced.
Essential Question	How do real and complex zeros together describe a polynomial, and how do you use that structure to answer a real-world model?
Materials	Pencils; student packet; calculator (optional). Desmos allowed for Try-It 3 check only, NOT for Practice #27.

[PIN] PRE-FLIGHT — P3 IS THE DOK-3 SPINE DAY

Today's conceptual core is Practice #27 (Storage Box). It is the single DOK-3 spine and it rehearses Topic 3 Assessment Q#6 (Lucy's tray).
Pace: Try-It 3a/3b (∼12 min) → Practice #17 (∼6 min) → Practice #27 (20 min). Keep Launch tight so Explore gets the full 38 min.
Desmos is allowed for Try-It 3 check only. Do NOT use it for Practice #27.
If running tight: follow the Period F cuts below. Never cut Practice #27.

Tag: bridge from Tuesday

Do Now **DOK:** 1–2 **Minutes:** 5

Teacher does

- Hand out at door.
- Post bridge prompt on board.
- Circulate 3 min; cold-call 1 student for one intersection point.

Students do

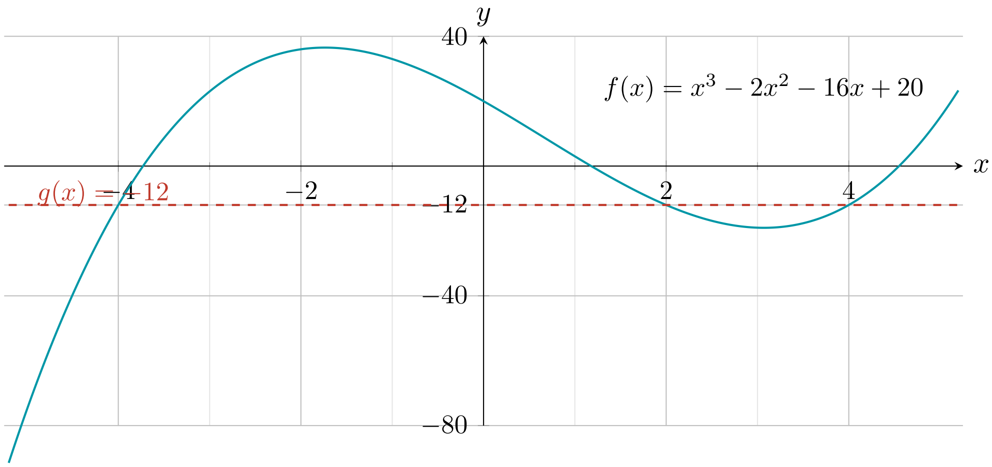
- List the intersection points.
- Respond to the bridge prompt verbally if called on.

Questions to ask

- How did you find where the graphs cross?
- Bridge: nature of the cubic's coefficients — why? (Conjugate irrational pair → rational coefficients.)

Adult role
Solo teacher: stand at door, circulate 3 min, cold-call.

Do Now (3-5-savvas-q11)
At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?



[CHECK] ANSWER KEY

Answer key (registry): Solve $x^3-2x^2-16x+20 = -12 \to x^3-2x^2-16x+32 = 0 \to (x-2)(x+4)(x-4) = 0$. Intersection points: $(-4,-12)$, $(2,-12)$, $(4,-12)$.

[LINK] BRIDGE FROM TUESDAY (verbal, posted on board)

A cubic has zeros 2, $1+\sqrt{3}$, and $1-\sqrt{3}$. What is the nature of its coefficients — rational, irrational, or complex? Why? (This is Topic 3 Assessment Q#2 — no surprises on Thursday.)

[LINK] BRIDGE PROMPT — Expected Answer

"Rational coefficients. The zeros $1+\sqrt{3}$ and $1-\sqrt{3}$ are irrational conjugates — when multiplied, $(x-(1+\sqrt{3}))(x-(1-\sqrt{3})) = x^2-2x-2$, which has rational coefficients. Multiplied by $(x-2)$, the full cubic has all rational coefficients."
This is Topic 3 Assessment Q#2, answer C.

[BOOK] COMPLEX-ZERO RULES (keep printed on your page)

- A polynomial with real coefficients that has complex zeros has them in CONJUGATE PAIRS.
- If $a+bi$ is a zero, then $a-bi$ must also be a zero.
- A degree-n polynomial has exactly n zeros (counting multiplicity). Some may be real, some complex.
- Example: if a cubic has a real zero at $x=4$ and a complex zero $2+3i$, its third zero must be $2-3i$.

[WARN] MODELING RULES (for the Storage Box)

- Set up a factored-form volume function $V(x) = (\text{expr}_1)(\text{expr}_2)(\text{expr}_3)$.
- Each factor is a physical dimension — label each: length, width, height.
- Use the physical constraint (e.g., "length is greatest") to decide which factor is which dimension.
- To find dimensions at $V = 10\ \text{ft}^3$, solve the cubic equation — show all work.

Tag: teacher models Example 3 + brief Example 4

Launch **DOK:** 2 **Minutes:** 12

Teacher does

- Project Example 3 (real + complex zeros cubic). Think aloud: find one real zero, divide, solve the quadratic.
- Brief Example 4 (modeling): 90 sec walk-through of Savvas's storage box — just the setup, not the solution.
- Pause once for 30-sec partner-check on conjugate pair.

Students do

- Watch silently through Example 3.
- Turn-and-talk: "Why does the quadratic produce the complex pair?"
- Write the complex-zero rule in packet margin.

Questions to ask

- If one real zero is $x = 4$, what's the next step?
- Why does the discriminant being negative produce complex zeros?
- How do you know the complex zero's conjugate is also a zero?

Adult role
Project. Think aloud. Do NOT solve Try-Its. Release promptly at minute 17.

[MIC] TEACHER SCRIPT

1. "Watch me find all zeros of this cubic. Step 1: guess a real zero by the rational root theorem. Step 2: synthetic divide. Step 3: solve the leftover quadratic."
2. When quadratic's discriminant is negative → "Now we get complex zeros. They come in conjugate pairs — so if I get $2 + 3i$, I also have $2 - 3i$. That's not a coincidence; it's a rule."
3. Example 4 flash: project the storage box picture. "Notice the factored form $V(x) = x(x+3)(x-1)$. We'll unpack this in Explore on the Practice #27 item."
4. Release: "You have Try-It 3a, 3b, then Practice #17, then the big one — #27. Manage your time."

Tag: Think-Pair-Share + DOK-3 modeling

Explore **DOK:** 2–3 **Minutes:** 38

Teacher does

- 5 laps of circulation across 38 min.

- Lap 1: Try-Its 3a/3b. Lap 2: Practice #17. Laps 3–5: Practice #27 ([STAR]).
- Do not give #27 answers. Pairs present parts (a)–(d) to each other; then you pick one pair for Share.

Students do

- Pairs work through Try-It 3a/3b (∼12 min), then #17 (∼6 min), then the full 20 min on #27.
- For #27, write each factor's physical meaning BEFORE solving part (d).
- Use sentence frame for the Share.

Questions to ask

- What are ALL zeros of this polynomial — real and complex?
- How do the factors of $V(x)$ relate to the box's physical dimensions?
- What does the constraint "length is greatest" tell you?
- For $V = 10$ ft^3, what cubic equation must you solve, and how?

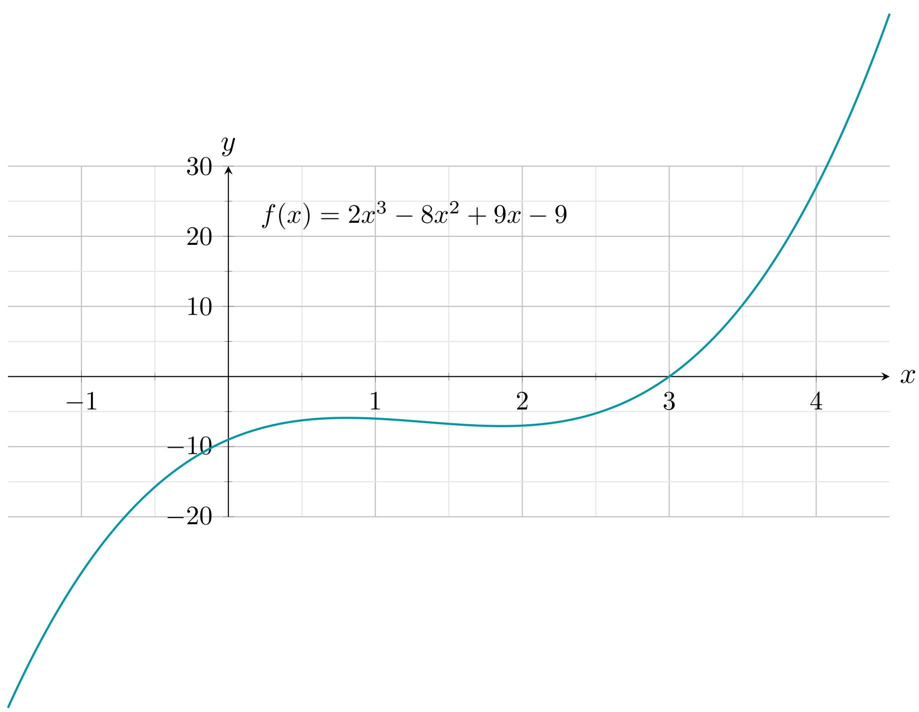
Adult role

Solo teacher: 5 laps of circulation. Press pairs on #27 part (c) — physical dimension interpretation is the DOK-3 pivot.

Work with a partner. Use the rule cards above. Storage Box (Practice #27) is the big one — plan 20 min for it.

Try It 3a (3-5-tryit-3a)

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = 2x^3 - 8x^2 + 9x - 9$.

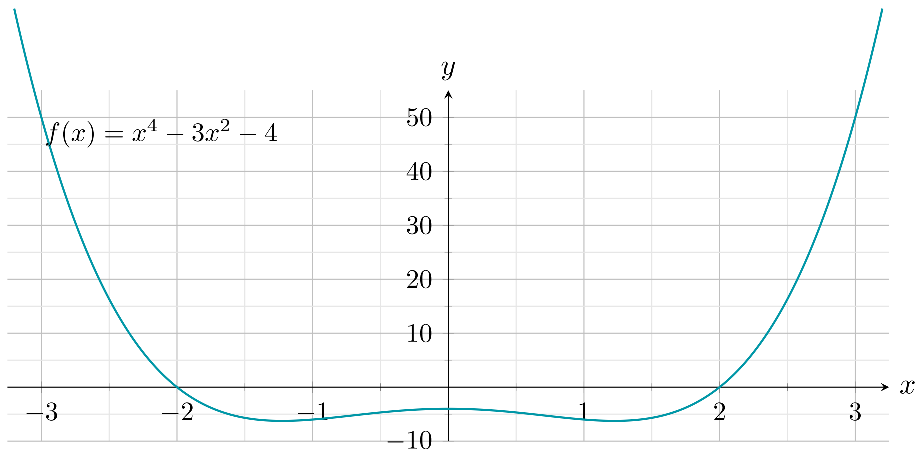


[CHECK] ANSWER / ANTICIPATED TALK

Answer key (registry): $x = 3$ (real, from graph + synthetic division), $x = \dfrac{1+i\sqrt{5}}{2}$, $x = \dfrac{1-i\sqrt{5}}{2}$ (complex, from quadratic formula on the depressed quadratic).

Try It 3b (3-5-tryit-3b)

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = x^4 - 3x^2 - 4$.

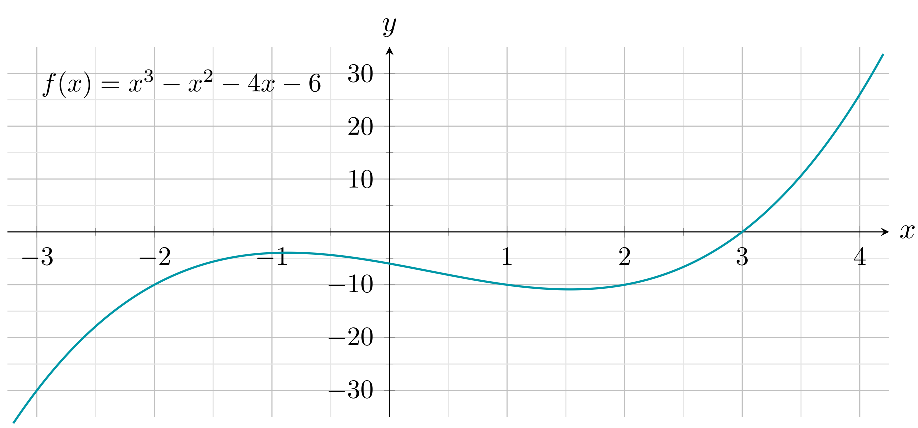


[CHECK] ANSWER / ANTICIPATED TALK

Answer key (registry): Factor as $(x^2-4)(x^2+1) = (x-2)(x+2)(x^2+1)$. Real zeros: $x = \pm 2$. Complex zeros: $x = \pm i$.

Practice #17 (3-5-savvas-q17)

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = x^3 - x^2 - 4x - 6$.



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (registry): Zeros: $x = 3$ (real, from graph), $x = -1+i$, $x = -1-i$ (from synthetic division by $(x-3)$ then quadratic formula on x^2+2x+2).

[SPEAK] CHAIN 1 CALLBACK — Storage Box (PLANT, April)

When to say: After students solve Practice #27, before Share. This plants the year-long extract-hidden-variable archetype — callbacks will land in L41, L51, L63, L64, L65.

"Notice what we just did: we had a physical object, we had one constraint (volume equals something), and we had to pull out a hidden dimension. That move — **build a polynomial from a constraint, then extract a dimension** — is a year-long move. We're going to see it again with a milk container in May, with a money formula in June. Today's work is how we learn the machine. Same move, new machine each time."

Practice #27 [STAR] DOK-3 DRIVER (3-5-savvas-q27)

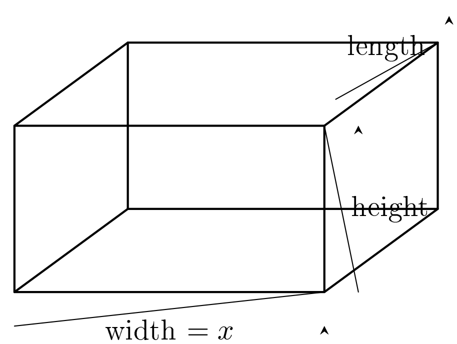
Model With Mathematics: The height of a rectangular storage box is less than both its length and width. The function $f(x) = x^3 + 2x^2 - 3x$ represents the volume of the rectangular box, where x represents the width of the box, in feet.

(a) Find the factored form of $f(x)$.

(b) Find the zeros of the function.

(c) You know x represents the width of the box. What do the other two factors represent?

(d) Find the dimensions of the box when the volume is 10 ft^3.



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (registry):
(a) $x(x^2 + 2x - 3) = x(x + 3)(x - 1)$.
(b) Zeros: $x = 0$, $x = -3$, $x = 1$.
(c) $(x - 1)$ represents the height; $(x + 3)$ represents the length.
(d) width = 2 ft, height = 1 ft, length = 5 ft.

Tag: academic conversation + assessment preview

Share / Summary **DOK:** 2 **Minutes:** 7

Teacher does

- Call 1 pair to the board to present #27 parts (a)–(d).
- Add 30-sec connection: "Tomorrow's assessment Q#6 (Lucy's tray) uses the same factored-volume idea."
- Post the exit ticket cue.

Students do

- Presenting pair walks through each part with sentence frame.
- Class signals thumbs up/sideways after each part.
- Write the Lucy-tray preview in margin.

Questions to ask

- Summarize the modeling strategy in one sentence.
- What's different between #27 and tomorrow's Lucy-tray problem? (Different physical setup, same factoring strategy.)

Adult role
Cold-call a pair that struggled earlier (not the strongest). Hold others accountable to listen.

Sentence Frames
Pair share: "For the Storage Box, the factored form was ___. I chose ___ as the length because ___. To find $V = 10$ ft^3, I ___."
Whole class: one pair at random presents Part (a)–(b)–(c)–(d) of #27.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 3

Teacher does

- Collect at door.
- Scan responses to #3 (assessment prediction). Use tonight to adjust Thursday Do Now emphasis.

Students do

- Complete 3 sentence stems.
- Be honest about #2 — the hardest step tells us what to hit Thursday morning.

Questions to ask

- Did students name a specific step in #2?
- Did they name Topic 3 content (factoring, multiplicity, conjugate pairs, modeling) in #3?

Adult role
Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

Today I learned that ___.
For the Storage Box, the hardest step was ___.
Thursday's assessment will test ___.

[CLOCK] PERIOD F — 45-MIN SHORT VARIANT (Wed only)

Period F has 45 min on Wednesday. Apply these cuts:
- Launch: skip Example 4 brief (stay on Example 3). Still 12 min.
- Explore: 25 min max. DROP Try-It 3b and Practice #17. Keep Try-It 3a + Practice #27 only. #27 is non-negotiable (DOK-3 spine + Thursday-assessment prep).
- Share: 5 min (cut from 7). One pair presents #27 parts (a)+(b) only; (c)+(d) as HW hint.
- Exit: 3 min (unchanged).
- Optional reinforcement: announce but don't expect completion.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide Complex-Zero Rules as a half-sheet cut-out.
- For #27, provide a pre-labeled box sketch (length / width / height slots).
- Accept partial solutions on #27(d) with verbal explanation of which factor is which dimension.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (do not skip).
- Word cards: "conjugate" = pair that together gives real numbers; "imaginary" = contains $i = \sqrt{-1}$; "model" = equation that describes a real situation.
- Pair ELL students with English-dominant partners for Share.
- On #27 part (c), allow verbal responses in students' strongest language before writing.